Driver onboarding pilot evaluation

Performance, short-term retention and the trade-offs behind a good operational decision.

Python · KPI analysis · Difference-in-differences

2 × 2

Pilot vs comparison
Before vs after

100%

fictional data

What decision are we trying to inform?

A process change should be assessed against both primary outcomes and possible unintended consequences.

BUSINESS QUESTION

Can onboarding be made more predictable without losing route reliability, people or quality?

01 **Reliability**
Routes within plan

02 **Retention**
Active on day 14

03 **Guardrails**
Delivery quality & safety

A reproducible fictional scenario

No names, figures, files or findings from a real workplace are included.

24

invented sites

4,800

synthetic route records

1,200

synthetic driver starts

Two groups × two periods • 100 routes and 25 starts per site per period

Every input is generated from a fixed random seed; outputs can be rerun independently.

How the comparison works

A simple 2×2 design distinguishes the pilot's change from the contemporaneous comparison-group change.

	BEFORE	AFTER
Pilot	Pilot	Pilot
Comparison	Comparison	Comparison

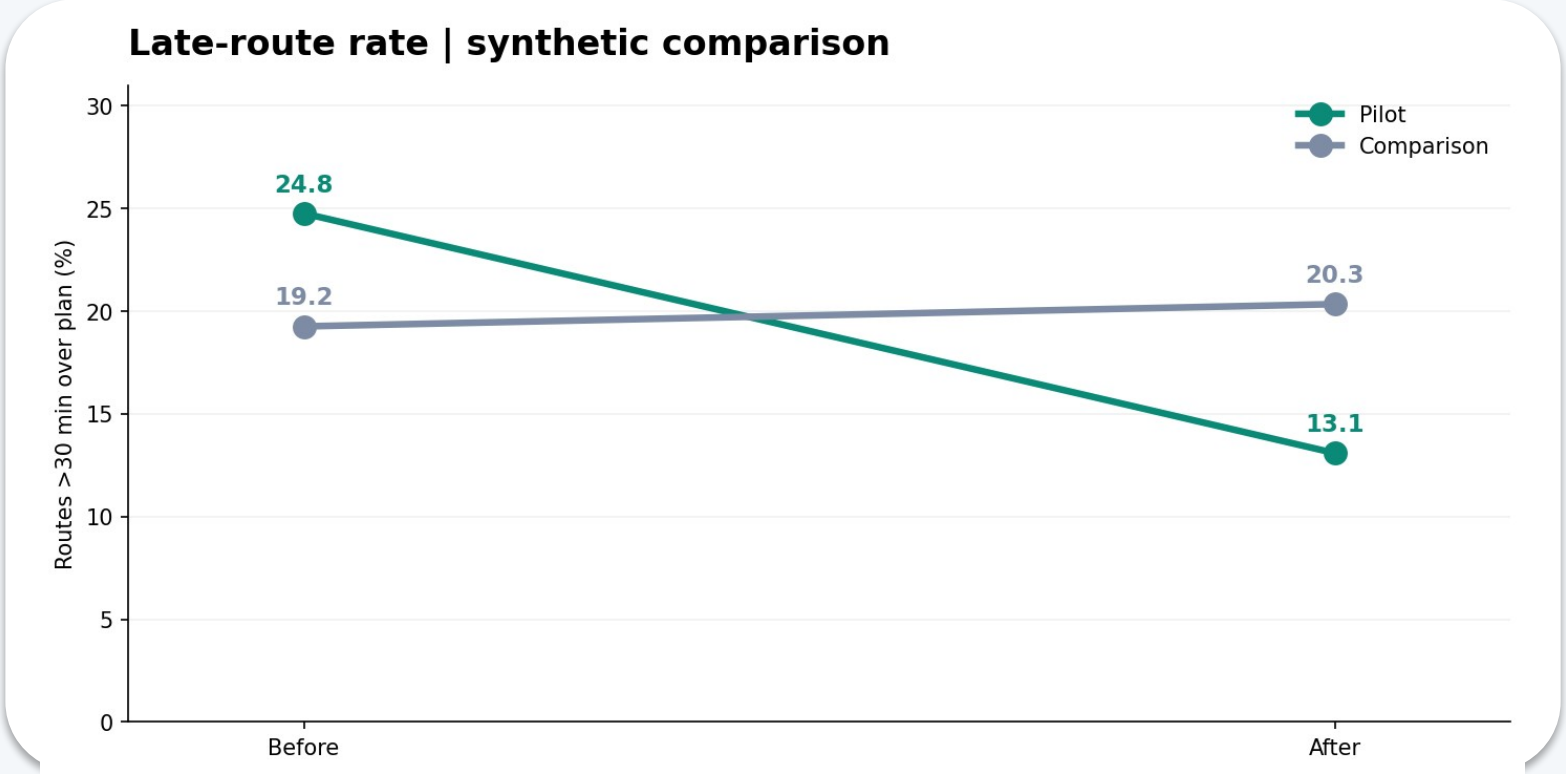
DIFFERENCE-IN-DIFFERENCES

Pilot change − Comparison change

Result is a relative change, not proof of causality.

Late-route rates improved in the fictional pilot

Primary KPI: share of routes more than 30 minutes beyond plan.



SYNTHETIC DID

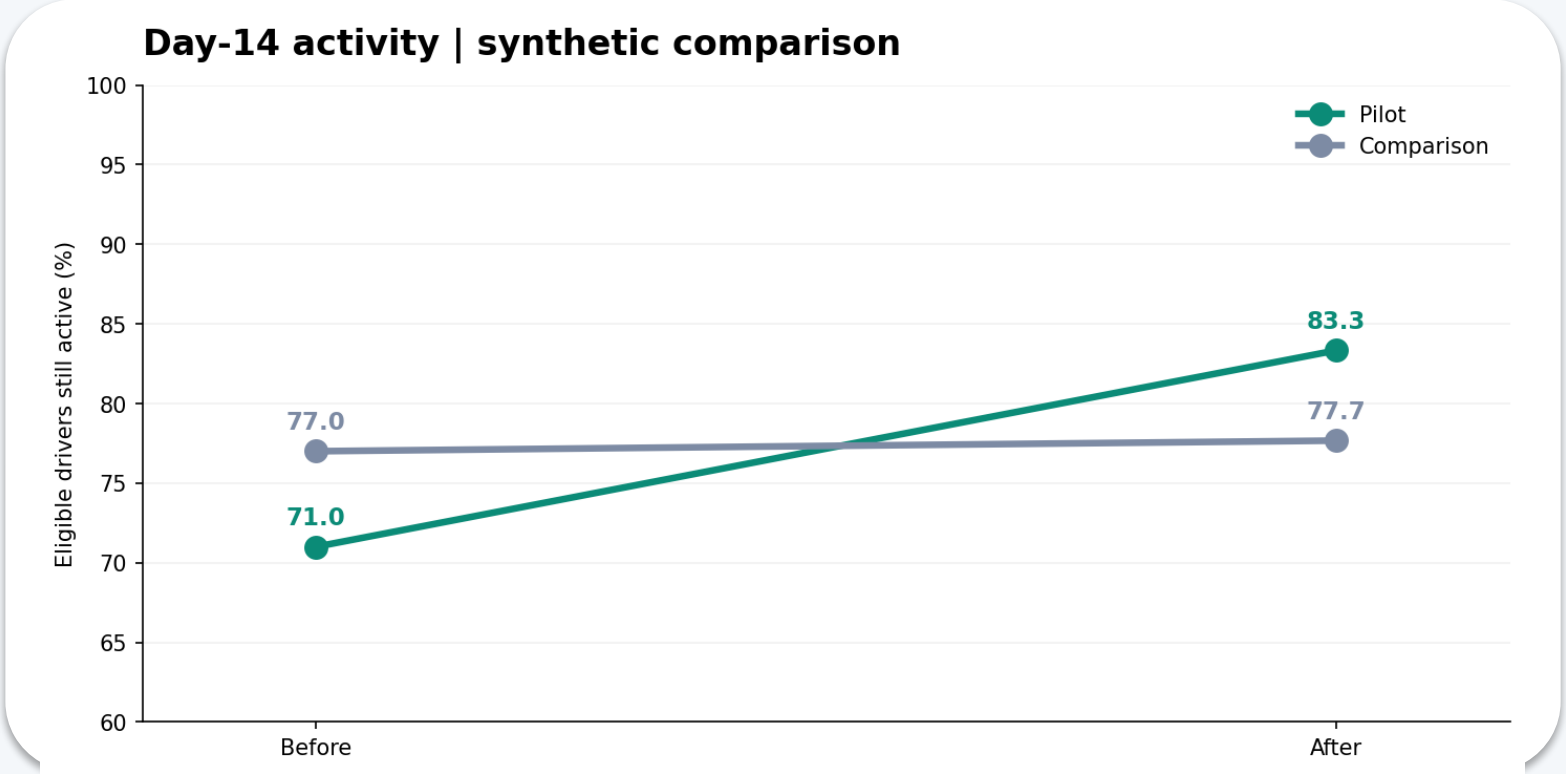
-12.75

percentage points

A favourable relative change is an analytical signal — not a verified intervention effect.

More fictional starters were active on day 14

The denominator includes eligible, matured new-driver cohorts only.



SYNTHETIC DID

+11.67

percentage points

Day-14 activity is a short-term indicator, not proof of skill or long-run retention.

Not every metric moved in the right direction

Primary gains should not obscure service-quality or safety concerns.

On-time dispatch

+2.75 pp

Favourable

Delivery completion

-0.68 pp

Unfavourable

Safety proxy

-14.07 pts

Unfavourable

Interpretation: monitor service and safety before recommending rollout.

Signals are not proof

Communicate analytical evidence without making unsupported claims.

WHAT THE FICTIONAL DATASET SHOWS

- Better relative route reliability
- Higher day-14 activity
- Improved dispatch
- Unfavourable quality/safety signals

WHAT IT CANNOT ESTABLISH

- Causation from just two periods
- Long-term retention or competence
- Actual safety incidents
- Realised monetary savings

What a responsible analyst would recommend

Better evidence matters more than a positive headline KPI.

01

Validate the comparison

Check route mix, site selection and whether group trends were similar before launch.

02

Investigate adverse guardrails

Break down completion and safety-proxy changes by site and driver tenure.

03

Strengthen retention analysis

Use consistently matured cohorts and later checkpoints, not only day 14.

04

Model costs only with approval

Use verified, authorised cost inputs and include downside scenarios.

Good analysis asks what changed — and why.

Reproducible Python scripts, transparent DiD calculations, short-term activity analysis and clear communication of operational trade-offs.

README / NOTEBOOK / CODE / SYNTHETIC DATA / SLIDES